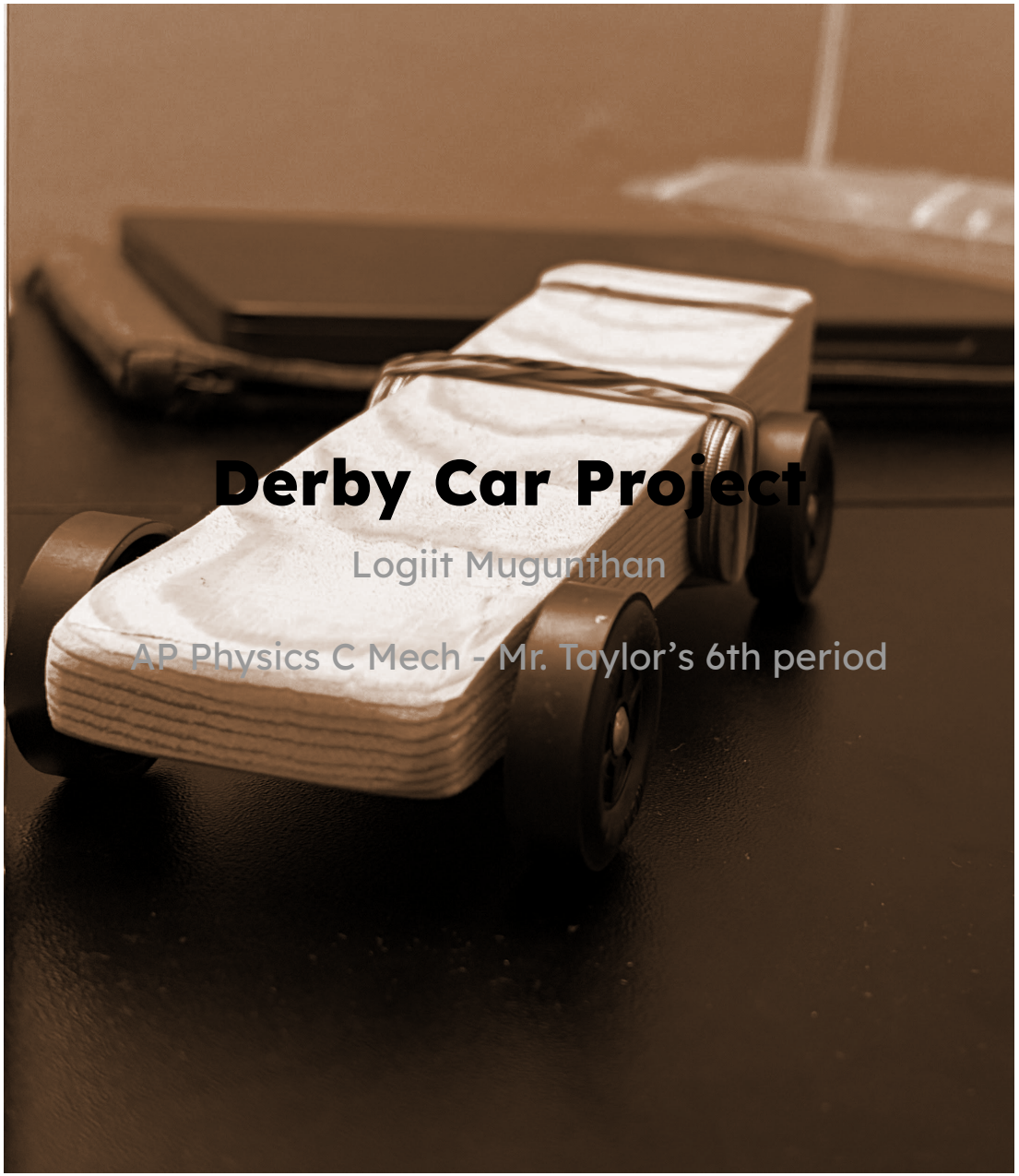


Derby Car Project

Logiit Mugunthan

AP Physics C Mech - Mr. Taylor’s 6th period

Stage 1

In stage 1 we each received mostly identical wooden blocks and 4 axles and 4 wheels. Then we took the initial time trials of just the block with the wheels on to have an established base case to compare our final design with. The following is my averaged time data for my wooden block. I had to take an average of 3 other people's results because I was absent on the day that the rest of the class took their time data. However, this should not have a great impact since all the blocks were mostly identical and the time data was mostly the same for the 3 people that I averaged the results from.

	Left Track	Right Track	Average
1	2.803 s	2.763 s	2.783 s
2	2.588 s	2.799 s	2.694 s
3	2.777 s	2.918 s	2.848 s

Mass of the block: 126.85
Time: 2.775 seconds

Next, we put the blocks through the wind tunnel to have initial data on the aerodynamics of the car. By testing just the block, we would be able to get a base case to which we can compare our final designs with. Due to time constraints and the fact that all the blocks were mostly identical, we decided to only put only one block through the wind tunnel and use that data for all the blocks.

Angle of Attack (AoA)	Maximum Lift (lbs)	Maximum Drag (lbs)	Lift/Drag Ratio

-5 degrees	0.06	0.35	0.17
0 degrees	0.01	0.34	0.03
5 degrees	0.13	0.36	0.36

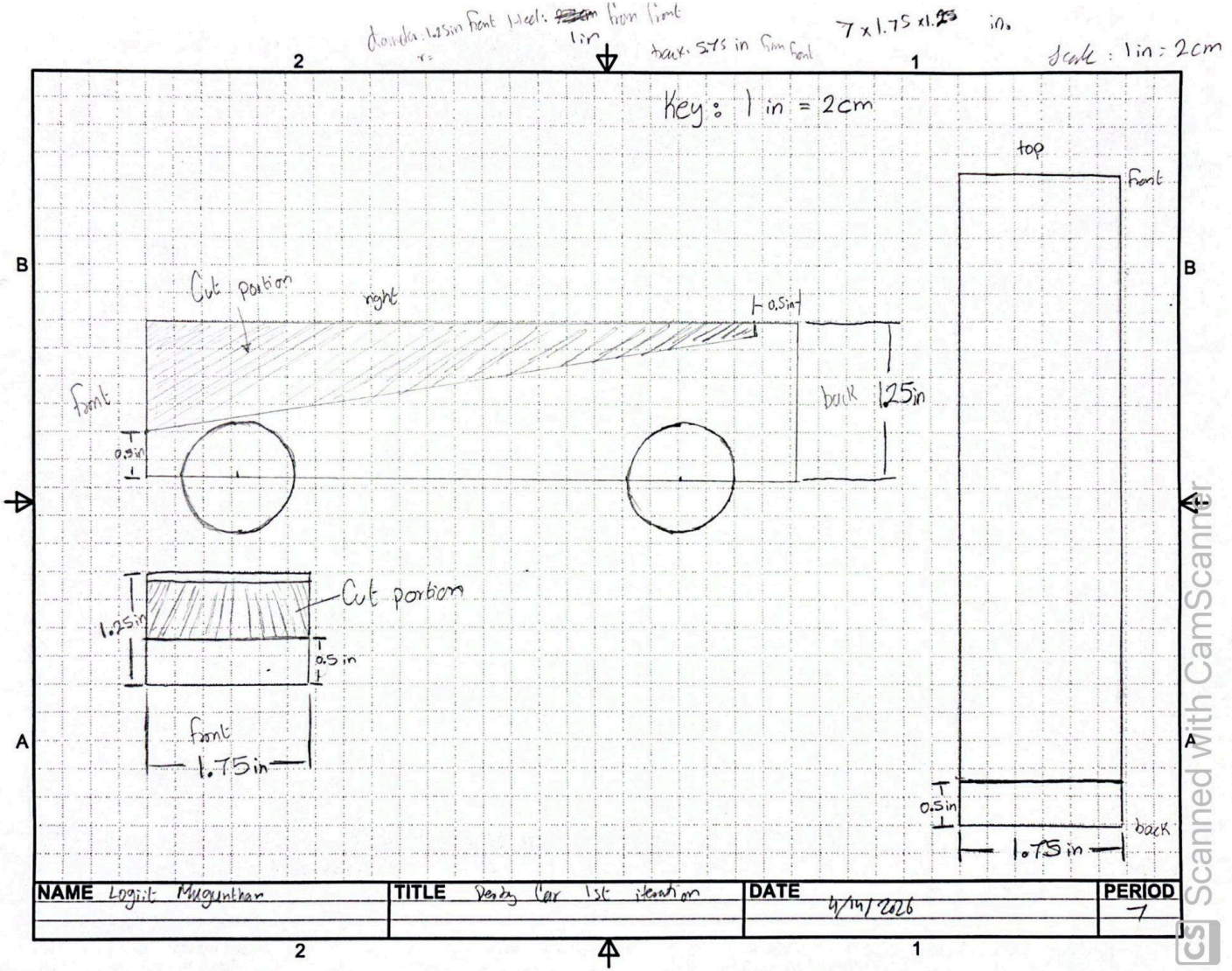
Overall, I found that there was very little lift and a lot of drag, which was to be expected since this was a solid block with no aerodynamic changes or modifications made to it. All the L/D ratio values were less than one which meant that the block was highly inefficient and had way more drag than lift. While the aim isn't to increase lift, I still need the L/D ratio to be 1 or slightly above 1 to ensure that drag is being minimized.

Stage 2

In stage 2 I focused on shaping the wood block to get my initial model of the car and to lube the axles to ensure that there would be negligible friction between the wheels and the axles. Reducing friction between those two would mean that there wouldn't be a loss of energy due to heat generated by the friction. While this energy lost through heat is marginal for a car of this scale, it still contributes to the overall speed of the car. By reducing friction, we also allow for the wheel to spin faster, for a longer period of time which increases the wheel's rotational velocity and directly increases the linear velocity of the car. Since the final goal is to have the fastest car, lubing the axles is a crucial step towards that. This effect was also observed when we ran the cars on the track again as the time for my car significantly decreased. In addition to lubing the axles, we also had to cut the car to shape. I decided on a wedge-style design for my car with most of the weight focused at the back of the car. I also decided to make what is usually considered the front of the car the back of mine since there was more mass above and behind the wheel at that spot. The total amount of mass removed from the cutting the sanding ended up being 33.46 grams. The reason that I wanted to keep as much mass as possible instead of making a big cut and taking off a lot of mass was because more mass would mean a higher initial potential energy for my car. Since we aren't launching the cars, but rather just letting them roll down the ramp, all that potential energy gets converted to kinetic energy by the time the car reaches the flat part of the ramp. A higher potential energy would mean a higher final kinetic energy and also a higher velocity. Another reason that I kept more mass was because with less mass, the car becomes more susceptible to other factors that would cause a loss of energy, such as friction caused by the air, or slight deviations within the track. Having more mass makes the car less susceptible to these factors and keeps more energy which means a higher final kinetic energy and higher velocity.

New mass: 93.39 grams
New average time: 2.499 seconds

Sketch of new car design:



Scanned with CamScanner

Pictures of new car design:







Stage 3

Due to time constraints this ended up being the final stage. In this stage I modified the mass of the car, making the car heavier. I aimed to add most of the mass near the rear axle of the car to bring its center of mass closer to the back of the car and thereby making it less resistant to small deviations during the run and also contributing to the fact that more initial mass means more initial potential energy being converted to kinetic energy. To add mass I used quarters since they are thin, small and are relatively massive compared to the other coins. I used 9 of these coins to add approximately 45 grams of mass. I placed 2 quarters on either side right in front of the rear wheels, one on the bottom and 3 at the very back of the car. This extra mass proved to be beneficial since the new time for the run decreased from the previous one by almost 0.02 seconds. While this was a small change, it was still an improvement on the previous time. Reflecting back, I think that I could have reduced more mass from the front of the car and added even more mass towards the back. That was how most of the cars that ended up going faster than mine were built and I think that having less mass in the front is more beneficial.

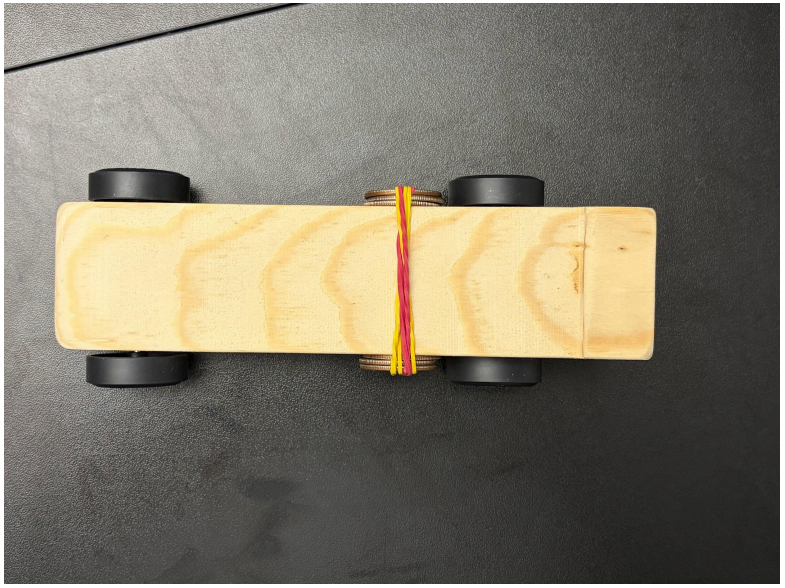
Final mass: 145.58 grams
Final time: 2.470 seconds

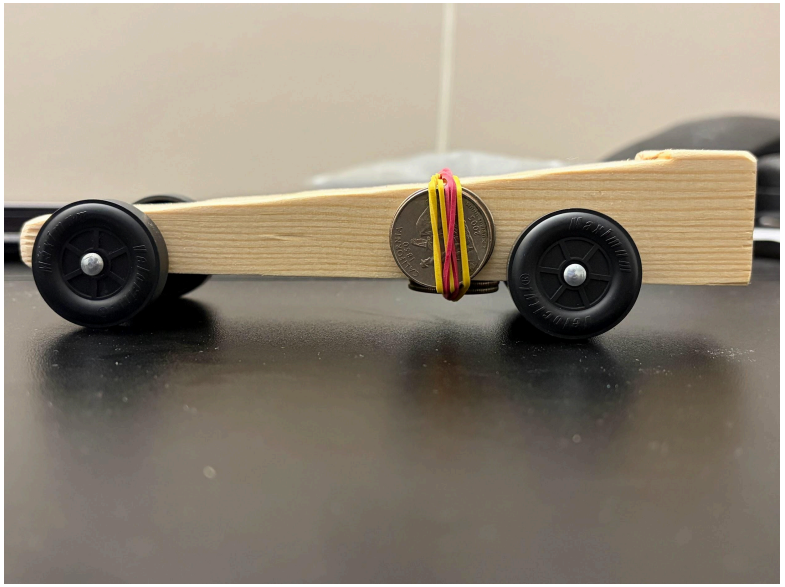
List of all changes made:

1. Dry lubricant on wheels to reduce friction between the wheel and the axle. This improves the speed of the car by reducing the energy lost to friction
2. A long diagonal cut almost all the way through the top of the car. Abruptly end the diagonal to leave a small ledge. Cutting the car to shape reduces the mass, which makes the car slower, but makes up for that in aerodynamic gain. The aerodynamic shape of the car reduces air resistance and hence reduces the energy lost to friction with the air, making the car go faster.
3. Adding roughly 45 grams of mass near the rear axles. Adding more mass ensures that the car is more stable and also contributes to the car having more initial potential energy. The mass placement towards the back of the car, near the rear axles, ensures that the center of mass of the car is located as far away from the front axles as possible to increase speed and also increase initial potential energy. This change also resulted in a faster car.

Pictures of the final car:







Reflection

Overall, I found the project to be fun and highly educational on the practical applications of both aerodynamics and energy. The changes that I made throughout the project each yielded a positive impact to the speed of the car with the shape and mass proving especially impactful. However, I did miss out on the aerodynamic testing in the wind tunnel after stage 2 to see how big of an impact the new shape had on the drag produced. Depending on the results of that test I might have been able to change the shape of the car slightly to minimize drag as much as possible. Instead of making the top of the car one smooth, slanted, singular surface, I opted to make another surface after the slant in an attempt to minimize drag and maximize aerodynamic potential. However, I never got to know whether that idea was successful because I wasn't able to put it through the wind tunnel to test since I was in a different period.